## MICROCONTROLEUR

### MCU
STM32F405RGT6

U2
+3V3
VBAT 1, VDD 19, VDD 32, VDD 48, VDD 64, VDD 13, VDDA
NRST 7
BOOT0 60
HSE_IN 5 PH0
HSE_OUT 6 PH1
PD2 54
PC0 8, PC1 9, PC2 10
PMS_RST 11 PC3
LORA_IO0 24 PC4
LORA_IO1 25 PC5
GPS_USART6_TX 37 PC6
GPS_USART6_RX 38 PC7
LED_BLUE 39 PC8
LED_GREEN 40 PC9
BARO_SPI3_SCK 51 PC10
BARO_SPI3_MISO 52 PC11
BARO_SPI3_MOSI 53 PC12
PC13 2
LSE_IN 3 PC14
LSE_OUT 4 PC15
VCAP_1 31
C11 2u2 GND
VCAP_2 47
C12 2u2 GND
VSS 18, VSSA 12, GND
PA0 14 UART4_TX
PA1 15 UART4_RX
PA2 16 PMS_SET
PA3 17 LORA_RESET
PA4 20 LORA_SPI1_NSS
PA5 21 LORA_SPI1_SCK
PA6 22 LORA_SPI1_MISO
PA7 23 LORA_SPI1_MOSI
PA8 41 LED_RED
PA9 42 USB_VBUS
PA10 43
PA11 44 USB_D-
PA12 45 USB_D+
PA13 46 JTMS-SWDIO
PA14 49 JTCK-SWCLK
PA15 50 JTDI
PB0 26 ONEWIRE
PB1 27 BARO_NCS
PB2 28
PB3 55 JTDO-SWO
PB4 56 JTRST
PB5 57 BUZZER
PB6 58 VCP_USART1_TX
PB7 59 VCP_USART1_RX
PB8 61 GPS_I2C1_SCL
PB9 62 GPS_I2C1_SDA
PB10 29
PB11 30
PB12 33 SAFEBOOT_GPS
PB13 34 EXTINT_GPS
PB14 35 PPS_GPS
PB15 36 RESET_GPS

### VBUS Sensing
VBUS Sensing tiré de AN4879

VBUS Sensing GPIO doit être configuré en entrée avec pull-up

VBUS_5V_USB
R1 10K
USB_VBUS R4 200 D1
Q1 2N7002
R2 10K
GND

### Connecteur USB
Résistance de tirage pour USB-D+ intégrée au STM32, pas besoin de l'ajouter en externe (cf AN4879).

VBUS_5V_USB
C1 4u7 GND
J1 USB4085-GF-A
A4 VBUS
R3 5k1 A5 CC1
R5 5k1 B5 CC2
USB_D- A7 D-
B7 D-
USB_D+ A6 D+
B6 D+
SBU1, SBU2, GND, SHIELD
A8, B8, A1, SH
GND

### Connecteur VCP
VCP_USART1_TX 1
VCP_USART1_RX 2
3
4
J2 0022053041
GND

### Découplage MCU
+3V3
C5 100nF, C6 100nF, C7 100nF, C8 100nF, C9 100nF, C10 100nF
GND

### Jumper BOOT0
+3V3
R20 10K
J3
BOOT0
GND

### Connecteur vers STLINK
J5 FTSH-107-01-L-DV-K-A
+3V3
GND
VCP_USART1_RX
JTMS-SWDIO
JTCK-SWCLK
JTDO-SWO
JTDI
JTRST
VCP_USART1_TX

### Bouton reset MCU
SW1 1825910-6
NRST
GND

### Cristaux
XTAL1 ECS-120-8-33-JGN-TR
HSE_IN
C15 GND
GND
HSE_OUT
C17 GND
Y1 ECS-.327-6-34RR-TR
LSE_IN
C18 GND
LSE_OUT
C19 GND

### LED RGB
LED1 150141M173100
LED_RED R24 200
LED_GREEN R25 200
K_R COMMON_A
K_G_ K_B
R26 200 LED_BLUE
+3V3
C16 100nF GND

## TELEMETRIE

### GPS
+3V3
D2
R7 3K3
BT1
C4 100nF
BACKUP_GPS
GND

U1 MAX-M10S-00B
+3V3
C2 10uF GND
C3 100nF GND
VCC 8, VCC_IO 7, V_BCKP 6
BACKUP_GPS R6 0
SAFEBOOT_GPS R8 0
SAFEBOOT 18
VIO_SEL 15
RESET_GPS R12 100
RESET R13 0 9
EXTINT_GPS R16 0 5 EXTINT
PPS_GPS R18 0 4 TIMEPULSE
R19 200
D3 LED_PPS
GND, GND, GND
19, 1, 12
GND
RF_IN 11
LNA_EN 13
VCC_RF 14
SCL 17
SDA 16
RXD 3
TXD 2
SMA pour YEGT002BA
AE1 Antenna_Shield
GND
+3V3
R9 22 P,Y
R10 22 P,Y
R11 0
R14 0
R15 0
R17 0 P,Y
GPS_I2C1_SCL
GPS_I2C1_SDA
GPS_USART6_TX
GPS_USART6_RX

### Module LoRa
RFM95W-868S2

U3
+3V3
LORA_SPI1_SCK 4 SCK
LORA_SPI1_MOSI 3 MOSI
LORA_SPI1_MISO 2 MISO
LORA_SPI1_NSS 5 NSS
LORA_RESET 6 RESET
3.3V 13
ANT 9
DIO5 7
DIO4 12
DIO3 11
DIO2 16
DIO1 15
DIO0 14
GND, GND, GND
1, 8, 10
GND
R21 100
R22 0 LORA_IO1
R23 0 LORA_IO0

U.Fl pour antenne patch
J4 CONUFL001-SMD-T
SIG 1
GND_2 3
GND_1 2
GND

+3V3
C13 100nF
GND

TP1 CLK LORA_SPI1_SCK
TP2 MOSI LORA_SPI1_MOSI
TP3 MISO LORA_SPI1_MISO
TP4 CS LORA_SPI1_NSS

## ALIMENTATION

TP5 ALIM
PWR_FLAG
J6
SW2 25139NAH
TP7 GND
GND

Protection des surtensions et des retours de courant
Q2 SQD50P04-13L_T4GE3
F1 1812L260_16MR
R27 400
20mA
D4 LED
R28 10K
C22 1uF
R30 2K
R29 0
GND
Q2 (P-MOSFET) : bloque les polarités inverses
F1 (POLYFUSE) : Protège des surintensités
D4 : Indique le statut de l'alimentation

Régulateur 3V3
IC1 LD1117DT33TR
IN 3, GND 1, OUT 2
C20 100nF
C21 10uF
TP6 3V3
+3V3

Régulateur 5V
IC3 L7805CDT-TR
IN 1, GND 2, OUT 3
C25 0,33uF
C26 100nF
TP8 5V
+5V

## EXPERIENCES

### Baromètre
IC2 KP466PXTMA1
BARO_NCS 1 NCS
BARO_SPI3_SCK 2 CLK
BARO_SPI3_MOSI 3 SDI
BARO_SPI3_MISO 4 SDO
R32 0 P,Y
R34 0 P,Y
GND 8, 7, 6, 5
+3V3
C23 100nF GND

### Capteur de température
U4 DS18B20+
+3V3
R31 4K7
GND 1
DQ 2 ONEWIRE
VDD 3
C24 100nF GND
+3V3

### Connecteur PMS5003 (Aerosols)
+3V3
R41 10K
GND
PMS_SET
UART4_TX
UART4_RX
PMS_RST
R33 0 P,Y
R35 0 P,Y
R40 10K
+3V3
+5V
J7 53048-0810
1 VCC
2 GND
3 SET
4 RX
5 TX
6 RST
7 NC
8 NC

## SIGNALETIQUES

### Buzzer
BUZZER
R36 100
R37 10K
+5V
D7
BZ1
NMOS_IRLZ34NPBF
Q3
GND

### Etat tensions
+3V3
R38 200
D5 LED
GND
+5V
R39 200
D6 LED
GND

H1 H2 H3 H4

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